The University of The Gambia

2026/27 – First Semester

Course Outline: Mathematics for Data Science (SDS 710)

Instructor: Dr. Malick SALLAH

Course Description

This course introduces the mathematical foundations required for modern data science. It combines key concepts from linear algebra and multivariable calculus with practical applications in optimization, modelling, and machine learning. Students will learn how matrices, derivatives, and numerical methods are used to analyse data, solve real-world problems, and build predictive models.

Course Objectives: By the end of this course, students should be able to

(i) Apply matrix methods to solve systems of linear equations and determine invertibility.
(ii) Compute determinants, eigenvalues, eigenvectors, and use diagonalization techniques.
(iii) Differentiate multivariable functions and Analyse gradients, Jacobians, and higher-order derivatives.
(iv) Solve unconstrained and constrained optimization problems using maxima/minima methods and Lagrange multipliers.
(v) Use linearisation and Taylor series to approximate multivariable functions.
(vi) Implement gradient descent and multilinear regression for practical data analysis problems.

Course Content

Invertible matrices, Systems of linear equations and Gaussian elimination, Determinants, Eigenvalues and eigenvectors, Matrix diagonalization, Partial differentiation and multivariate chain rule, Maxima and minima, Lagrange multipliers, Gradients of vector-valued functions and matrices, Higher order derivatives & Linearisation and multivariate Taylor series, Gradient descent and Multi-linear Regression.

Laboratory Work

Students will undertake practical sessions using computational tools such as Python, MATLAB, MATHEMATICA, or R to reinforce theoretical concepts. Laboratory exercises may include:

- Solving matrix equations numerically Computing eigenvalues and eigenvectors using software
- Visualizing multivariable functions and gradients
- Implementing optimization algorithms such as gradient descent
- Performing multilinear regression on real datasets
- Applying Taylor approximations in numerical analysis
- Mini-projects involving predictive modelling and data interpretation

Course Materials / Equipment

- Lecture notes and recommended textbooks
- Scientific calculator
- Laptop or desktop computer
- Internet access
- Installed software: Python / MATLAB / MATHEMATICA / R
- Access to datasets for laboratory practice
- Whiteboard and projector for instruction

Delivery Method

The course will be delivered through a blend of:

- Face-to-face lectures
- Tutorials and guided problem-solving sessions
- Computer laboratory practical
- Group discussions and presentations

- Online learning resources and assignments
- Independent study and project-based learning

Assessment

Student performance will be evaluated through theoretical understanding and practical application. Assessment will be continuous and summative.

- Continuous Assessment (50% of total mark): This is understood to include Tests, Homework and quizzes.
- Final Examination (50% of total mark): Formal Written Examination (theory, derivations, and applied problems) – 50%
- The continuous assessment and final examination scores will be combined for a total of 100%. The standard UTG grading system will be applied.